

CollabSolve Jharkhand

AI-Powered Civic Intelligence, Multimodal Telephony & Autonomous Redressal Platform

TOTAL DURATION**5 Minutes (Strict)****PACING / WORD COUNT****670 Words (~135 wpm)****PRIMARY AI ENGINE****Gemini 2.5 Flash****LIVE REPO****github.com/jeevan123-craz**

5-MINUTE TIME ALLOCATION MATRIX

0:00–0:45: Ground Problem & The Hook | **0:45–1:30:** 4-Stakeholder Architecture | **1:30–2:30:** Live Demo: Multimodal Voice & GPS | **2:30–3:15:** Live Demo: Twilio Phone Telephony | **3:15–3:45:** Live Demo: Admin AI Clustering | **3:45–4:30:** Tech Stack & Security | **4:30–5:00:** Impact & Closing

1. The Hook & The Ground Problem of Jharkhand

0:00 – 0:45 (45 Seconds)

 **Screen Cue:** Display Home Page hero banner at `localhost:3000` or PPT Title Slide. Stand tall, make direct eye contact with the jury, and speak with calm authority.

SPEAKER:

"Respected Jury Members and Evaluators, Namaskar.

In Jharkhand, over **70% of our population resides in rural and semi-urban districts**. When a drinking water pipeline breaks in Dumka, or a community healthcare center runs out of life-saving antivenom in West Singhbhum — **how does that ground reality reach administrative decision-makers?**

Today, citizens face an impossible digital divide: complex form fields, English-only websites, and opaque complaint registries. At the same time, university researchers who want to engineer solutions have zero access to ground data, while nodal officers are paralyzed by hundreds of duplicate, unstructured complaints.

We present **CollabSolve Jharkhand** — an AI-powered civic intelligence, crowdsourced R&D, and autonomous redressal platform that transforms ordinary citizen voices into measurable, accountable governance."

2. The Solution Architecture: The 4-Way Ecosystem

0:45 – 1:30 (45 Seconds)

 **Screen Cue:** Switch to System Architecture Diagram or Platform Overview showing the 4 roles.

SPEAKER:

"CollabSolve bridges this gap by unifying four key stakeholders into a single automated lifecycle:

- 1. Citizens:** Who report grievances hands-free in their natural voice without typing.
- 2. Administrative Nodal Officers:** Who receive AI-clustered, deduplicated themes and instant executive summaries.
- 3. University Researchers:** Who convert recurring civic problems into capstone R&D challenges with pre-screened feasibility scores.
- 4. Industry CSR Partners:** Who fund and adopt viable solutions for state-wide deployment.

Let us demonstrate this live in action."

3. LIVE DEMO 1: Multimodal AI Voice & Auto-Geolocation

1:30 – 2:30 (60 Seconds)

 **Screen Action:** Open `localhost:3000/citizen/voice`. Click **'Start Conversation'**. Speak clearly into the microphone: "A major water pipeline has burst near the main road in Ranchi and dirty water is flooding homes."

SPEAKER:

"Imagine a rural citizen who cannot read or write English. On CollabSolve, they simply tap **'Voice Mode'**.

Powered by **Gemini 2.5 Flash**, our assistant doesn't just record audio — it engages in an empathetic, multi-turn dialogue with intelligent counter-questioning.

Notice two technical breakthroughs right here on the screen:

- **Automated Geolocation Precision:** Through browser GPS and reverse geocoding via OpenStreetMap, the platform automatically detects their exact district — Ranchi — without requiring manual selection.
- **Real-Time Live Form Synthesis:** As the citizen speaks, Gemini parses the raw speech, creates an administrative Title, categorizes it under 'Water Supply', and auto-populates the official grievance form in real time.

The citizen speaks naturally, and the platform handles the bureaucratic paperwork."

4. LIVE DEMO 2: Autonomous Telephony via Twilio AI

2:30 – 3:15 (45 Seconds)

 **Screen Action:** Highlight the `/api/twilio/call` endpoint and telephony architecture, or show call dispatch trigger in Admin Kanban.

SPEAKER:

"Now, what about citizens in remote forest villages with basic 2G feature phones and zero internet access?

We engineered **Twilio Autonomous AI Telephony**. When an issue is resolved on the ground, or when the nodal department requires on-site verification, our system automatically places an outbound voice call:

'Namaskar! This is an automated update from CollabSolve Jharkhand. Your reported challenge regarding water supply in Ranchi has been resolved.'

The citizen can even speak their update over the standard phone call. Gemini transcribes and logs their verbal response directly into the official audit trail. **This delivers true universal access.**"

5. LIVE DEMO 3: Administrative AI Clustering & Command Center

3:15 – 3:45 (30 Seconds)

 **Screen Action:** Open `localhost:3000/admin`. Toggle view to '**AI Clusters**' and click '**Run AI Triage**'. Show clustered theme cards.

SPEAKER:

"Let us now look at the Nodal Officer's Command Center.

In traditional systems, if 30 citizens report drainage issues in Dhanbad, the officer gets 30 separate tickets.

With one click on '**Run AI Triage**', our clustering algorithm groups duplicate reports into a unified **Theme**, computes an Urgency score from Critical to Low, and assesses citizen sentiment.

With '**Generate AI Digest**', the admin receives an auto-drafted, one-page executive brief formatted for direct submission to ministry secretaries."

6. Researcher AI Quality Checker & Tech Architecture

3:45 – 4:30 (45 Seconds)

 **Screen Action:** Show Challenge detail page /chaLLenges/1 with 'AI Proposal Review' button and 100-point score meter.

SPEAKER:

"When academic researchers propose technical solutions, our built-in **AI Proposal Reviewer** evaluates their pitch against the original challenge description, scoring it out of 100 on Feasibility, Budget Realism, and Technical Clarity before human review.

Our Production Architecture:

- **Frontend & Edge:** Next.js 16 with React 19 and Turbopack.
- **Database & Realtime:** Supabase PostgreSQL with WebSocket channels.
- **AI Layer:** Google Gemini 2.5 Flash for sub-second multimodal reasoning.
- **Telephony:** Twilio Voice with TwiML speech streaming.
- Every line of code compiles with zero errors and is live on GitHub."

7. Measurable Impact & The Winning Closing Statement

4:30 – 5:00 (30 Seconds)

 **Screen Action:** Return to Main Home Page showing the Jharkhand Heatmap, or display Concluding Slide with Team Contact & GitHub link.

SPEAKER:

"CollabSolve Jharkhand is not just another grievance portal. It is an end-to-end civic R&D engine.

It turns isolated ground complaints into university innovation challenges, funded by CSR partners, and monitored transparently by the state government.

By collapsing the barrier of reporting to a simple voice conversation, and closing the resolution loop with automated phone calls, we ensure that in the state of Jharkhand: **No citizen is left unheard, and no challenge is left unsolved.**

Thank you, and we look forward to your questions."

Q1: "How does your system handle tribal languages like Santhali, Mundari, or regional dialects like Khortha?"

Answer: "Our audio pipeline routes acoustic input through multilingual speech models and passes the phonetic text to Gemini 2.5 Flash with few-shot Jharkhand dialect mappings. The AI translates regional idioms into standardized Hindi and English administrative records while preserving the exact ground context."

Q2: "What prevents malicious users from flooding fake issues to game the system?"

Answer: "We employ three layers of verification: First, semantic duplicate detection (`/api/ai/check-duplicate`) identifies coordinated bot spam. Second, GPS reverse geocoding verifies spatial clustering. Third, sentiment and urgency scoring flags inconsistent or fabricated descriptions for mandatory manual review."

Q3: "How scalable is the Twilio phone calling feature for thousands of users?"

Answer: "The telephony engine is decoupled into serverless webhook queues. Urgent alerts (e.g., public health hazards) are triggered instantly via priority queues, while routine resolution notifications are throttled during off-peak hours to respect telecom rate limits and optimize operational costs."

Q4: "Is this platform actually built, or is this just a mockup?"

Answer: "It is completely functional and production-ready. All 43 routes pass next build with zero TypeScript errors. The entire repository is published on GitHub at github.com/jeevan123-craz/collabsolve-jharkhand with Supabase realtime connectivity and Gemini API integration."